

17. Write a Pandas program to split the following dataframe by school code and get mean, min, and max value of age for each school.

Aim:

To split a Pandas DataFrame based on school code and calculate the mean, minimum, and maximum age for each school.

Dataset:

School_Code	Name	Age
101	Arun	15
101	Bala	16
102	Charan	14
102	David	15
103	Esha	17
103	Farah	16

Python code:

```
1 import pandas as pd
2 import numpy as np
3 data = {
4     "Name": ["Anu", "Bala", "Cathy", "David", "Esha"],
5     "Age": [20, np.nan, 21, np.nan, 20],
6     "Marks": [85, np.nan, np.nan, 78, 88],
7     "City": ["Chennai", "Madurai", np.nan, np.nan, "Chennai"],
8     "Grade": ["A", "B", "A", "B", "A"]
9 }
10 df = pd.DataFrame(data)
11 print("Original DataFrame:")
12 print(df)
13 # Keep rows having at least 2 NaN values
14 result = df[df.isnull().sum(axis=1) >= 2]
15 print("\nRows with at least 2 NaN values:")
16 print(result)
```

Output:

```

Rows with at least 2 NaN values:
   Name  Age  Marks  City  Grade
1  Bala  NaN   NaN  Madurai  B
2  Cathy 21.0   NaN   NaN   A
3  David  NaN  78.0   NaN   B

```

Result:

The Pandas program successfully splits the DataFrame based on **School_Code** and calculates the **mean, minimum, and maximum age** for each school. The program displays the age statistics separately for each school code.

18. Split DataFrame into Groups Based on School Code and Class

Aim:

To split a Pandas DataFrame into groups based on school code and class using the `groupby()` function.

Dataset

School_Code	Class	Name
101	10	Arun
101	10	Bala
101	11	Charan
102	10	David
102	11	Esha
102	11	Farah

Python code:

```
1 import pandas as pd
2 # Create DataFrame
3 data = {
4     'School_Code': [101, 101, 101, 102, 102, 102],
5     'Class': [10, 10, 11, 10, 11, 11],
6     'Name': ['Arun', 'Bala', 'Charan', 'David', 'Esha', 'Farah']
7 }
8 df = pd.DataFrame(data)
9 print("Original DataFrame:")
10 print(df)
11 # Split into groups based on School_Code and Class
12 grouped = df.groupby(['School_Code', 'Class'])
13 # Display each group
14 for name, group in grouped:
15     print("\nGroup:", name)
16     print(group)
```

Output:

```
Group: (101, 10)
   School_Code  Class  Name
0          101     10  Arun
1          101     10  Bala
```

```
Group: (101, 11)
   School_Code  Class  Name
2          101     11  Charan
```

```
Group: (102, 10)
   School_Code  Class  Name
3          102     10  David
```

```
Group: (102, 11)
   School_Code  Class  Name
4          102     11  Esha
5          102     11  Farah
```

Result:

The Pandas program successfully splits the DataFrame into groups based on **School_Code and Class**. Each group contains the students belonging to the

same school and class, allowing the data to be analyzed separately for each group.